

JEE(Main + Advanced) : ENTHUSIAST & LEADER COURSE (SCORE ADVANCED)

ANSWER KEY

PAPER-1 (OPTIONAL)

PART-1 : PHYSICS

SECTION-I (i)	Q.	1	2	3	4		
	A.	A	C	D	C		
SECTION-I (ii)	Q.	5	6	7			
	A.	A,C	B,C	A,B,C			
SECTION-I (iii)	Q.	8	9	10	11		
	A.	D	C	A	C		
SECTION-II	Q.	1	2	3	4	5	6
	A.	594	4	3	1	1	22

PART-2 : CHEMISTRY

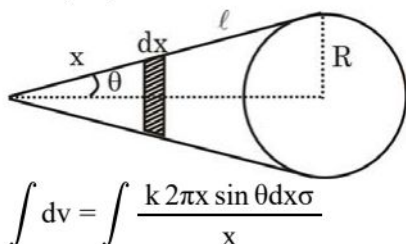
SECTION-I (i)	Q.	1	2	3	4		
	A.	A	B	D	D		
SECTION-I (ii)	Q.	5	6	7			
	A.	B,C	B,C	A,C			
SECTION-I (iii)	Q.	8	9	10	11		
	A.	C	A	B	B		
SECTION-II	Q.	1	2	3	4	5	6
	A.	4	160	3	331	3	2

PART-3 : MATHEMATICS

SECTION-I (i)	Q.	1	2	3	4		
	A.	B	C	C	C		
SECTION-I (ii)	Q.	5	6	7			
	A.	A,C	A,B,C,D	A,C			
SECTION-I (iii)	Q.	8	9	10	11		
	A.	D	B	A	B		
SECTION-II	Q.	1	2	3	4	5	6
	A.	75	3	1	6	1	0

HINT – SHEET
PART-1 : PHYSICS
SECTION-I (i)

1. Ans (A)



$$V = 2\pi\sigma k \sin \theta \ell = 2\pi\sigma k \frac{R}{\ell} \ell$$

$$= 2\pi\sigma R \times \frac{2}{4\pi\epsilon_0} = \frac{\sigma R}{2\epsilon_0}$$

$$V(\text{after removal of part AB}) = \frac{5}{6} \times \frac{\sigma R}{2\epsilon_0}$$

$$= \frac{5\sigma R}{12\epsilon_0} = \frac{5 \times R \times 12\epsilon_0}{12\epsilon_0} = 5V$$

2. Ans (C)

$$J \cdot \frac{a}{\sqrt{2}} = \frac{(2m)a^2}{12} \cdot \omega$$

$$\Rightarrow \omega = \frac{3\sqrt{2}}{ma} J$$

$$2T = m \cdot \omega^2 \cdot \frac{a}{3\sqrt{2}} \Rightarrow T = \frac{3J^2}{\sqrt{2}ma}$$

3. Ans (D)

$$\text{Gain in PE of ejected part} = (\rho a^3)g \left(\frac{a}{2}\right) \propto a^4$$

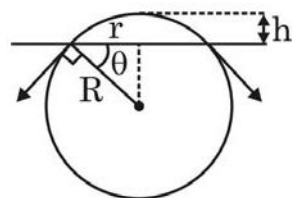
$$\text{KE of meteor} \propto a^4$$

$$a \propto (\text{KE})^{1/4}$$

$$\text{Volume} \propto a^3 \propto (\text{KE})^{3/4}$$

as mass doubles, KE doubles and volume becomes $2^{3/4}$ times.

4. Ans (C)



$$\left(\frac{V}{d}\right) \cdot \left(\frac{V \epsilon_0}{d} \pi r^2\right) \times \frac{1}{2} = \sigma \times 4\pi r \times \cos \theta$$

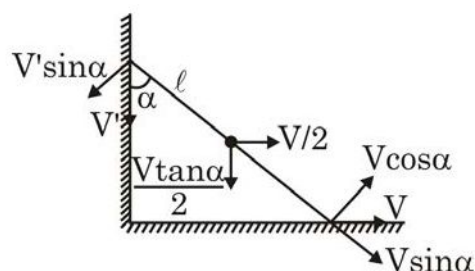
$$\Rightarrow \cos \theta = \frac{\epsilon_0 V^2 r}{8\sigma d^2}$$

$$h \approx \frac{1}{2} r \cos \theta = \frac{\epsilon_0 V^2 r^2}{16\sigma d^2}$$

PART-1 : PHYSICS

SECTION-I (ii)

5. Ans (A,C)



$$V \sin \alpha = V' \cos \alpha$$

$$V' = V \tan \alpha$$

$$\frac{d\alpha}{dt} = \frac{V \cos \alpha + V' \sin \alpha}{2l}$$

$$\frac{d\alpha}{dt} = \frac{V \cos \alpha + V \tan \alpha \sin \alpha}{2l}$$

$$\frac{d\alpha}{dt} = \frac{V}{2l \cos \alpha}$$

Horizontal acceleration of sphere = 0

$$\text{Vertical acceleration} = d \left(\frac{V \tan \alpha}{2} \right) / (dt) = \frac{V}{2} \sec^2 \alpha \frac{d\alpha}{dt}$$

$$= \frac{V}{2} \sec^2 \alpha \times \frac{V}{2l \cos \alpha}$$

$$\text{Vertical acceleration} = \frac{V^2}{4l} \times \frac{1}{\cos^3 \alpha}$$

$$mg - N = \frac{mV^2}{4l \cos^3 \alpha}$$

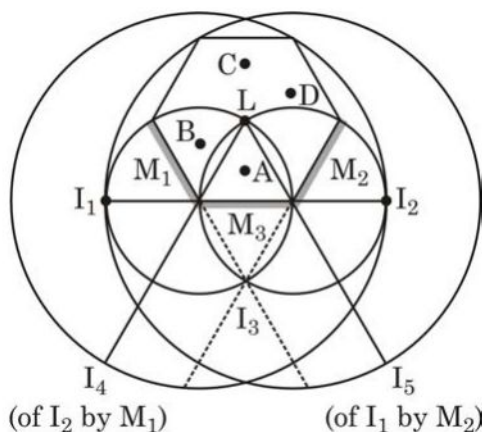
$$N = mg - \frac{mV^2}{4l \cos^3 \alpha}$$

$$= mg - \frac{mV^2 2\sqrt{2}}{4l} = mg - \frac{mV^2}{\sqrt{2}l}$$

$$= m \left[g - \frac{V^2}{\sqrt{2}l} \right]$$

$$\text{Vertical acceleration } (\alpha = 60) = \frac{V^2}{4l} 8 = \frac{2V^2}{l}$$

6. Ans (B,C)



(of I_2 by M_1) (of I_1 by M_2)

A lies in the field of view of I_1 , I_2 and I_3 , but

not of I_4 & I_5 .

B lies in the field of view of I_1 , I_2 , I_3 and I_4 , but

not of I_5 .

C lies in the field of view of all $I_1, I_2, I_3,$

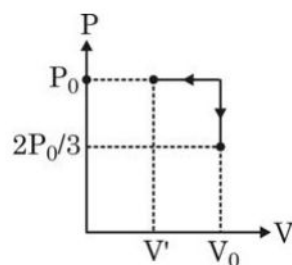
I_4 and I_5 .

D lies in the field of view of I_1, I_2, I_3 and I_5 but

not of I_4 .

7. Ans (A,B,C)

In 1st case, as the air enters slowly, there will be no change in its temperature (T_0). The air will fill until the pressure become $\frac{2P_0}{3}$. After releasing the piston (massless), due to outside and inside pressure difference, the piston suddenly falls down and the gas is compressed suddenly. The P-V graph of the process can be given as



$$\Delta Q = 0$$

$$\Rightarrow -P_0 \cdot (V_0 - V') + n \cdot \frac{5R}{2} (T - T_0) = 0$$

$$\Rightarrow -P_0 V_0 + P_0 V' + \frac{5}{2} nR(T - T_0) = 0$$

$$\Rightarrow -\frac{3}{2} nRT_0 + nRT + \frac{5}{2} nRT - \frac{5}{2} nRT_0 = 0$$

$$\Rightarrow \frac{7}{2} T = \frac{8}{2} T_0 \Rightarrow T = \frac{8}{7} T_0$$

In 2nd case, as the air enters through valve, the work done by atmospheric air on the entering air goes in increasing internal energy of entering air.

$$\Rightarrow P_0 \cdot \left(\frac{nRT_0}{P_0} \right) = n \cdot \frac{5R}{2} \cdot (T - T_0)$$

$$T_0 = \frac{5}{2} T - \frac{5}{2} T_0 \Rightarrow T = \frac{7}{5} T_0$$

Now, when the piston is released, final temperature

will become $\frac{8}{5} T_0$.

PART-1 : PHYSICS

SECTION-I (iii)

8. **Ans (D)**

$$(P) : \Delta x_1 = CP_1 \frac{d}{D} = 0.3 \mu\text{m} = \frac{3}{4} \lambda$$

$$\Delta x_2 = CP_2 \frac{d}{D} = 1.2 \mu\text{m} = 3\lambda$$

$$\Delta x_C = 0$$

$$\Rightarrow I_C = 4I_0, I_{P_1} = 2I_0, I_{P_2} = 4I_0$$

$$(Q) : \Delta x_C = 0 \Rightarrow I_C = 4I_0$$

$$\Delta x_1 = \mu CP_1 \frac{d}{D} = 0.4 \mu\text{m} = \lambda$$

$$I_{P_1} = 4I_0$$

$$\Delta x_2 = \mu CP_2 \frac{d}{D} = 1.6 \mu\text{m} = 4\lambda$$

$$I_{P_2} = 4I_0$$

$$(R) : \Delta x_C = (\mu - 1)t = 0.4 \mu\text{m} = \lambda$$

$$I_C = 4I_0$$

$$\Delta x_1 = 0.3 \mu\text{m} - 0.4 \mu\text{m} = -0.1 \mu\text{m} = \frac{-\lambda}{4}$$

$$\Rightarrow I_{P_1} = 2I_0$$

$$\Delta x_2 = 0.4 \mu\text{m} - 0.4 \mu\text{m} = 0$$

$$\Rightarrow I_{P_2} = 4I_0$$

$$(S) : \Delta x_C = d \sin \theta = d \theta = 0.1 \mu\text{m} = \frac{\lambda}{4}$$

$$\Rightarrow I_C = 2I_0$$

$$\Delta x_1 = 0.3 \mu\text{m} - 0.1 \mu\text{m} = 0.2 \mu\text{m} = \frac{\lambda}{2}$$

$$\Rightarrow I_{P_1} = 0$$

$$\Delta x_2 = 0.4 \mu\text{m} - 0.1 \mu\text{m} = 0.3 \mu\text{m} = \frac{3\lambda}{4}$$

$$\Rightarrow I_{P_2} = 3I_0$$

9. **Ans (C)**

$$mvr = \frac{nh}{2\pi}$$

$$\frac{mv^2}{r} = \frac{K}{r^2}$$

$$mv^2 r = K$$

$$v \propto \frac{1}{n}$$

$$r \propto n^2$$

$$\omega \propto \frac{1}{n^3}$$

$$T \propto n^3$$

10. **Ans (A)**

$$\epsilon_1 = \frac{x}{\ell}$$

$$\epsilon_2 = \frac{x}{4\ell}$$

$$\frac{\epsilon_1}{\epsilon_2} = 4$$

$$\sigma_1 = \sigma_2$$

$$\Rightarrow y_1 \epsilon_1 = y_2 \epsilon_2 \Rightarrow 4y_1 = y_2$$

$$u = \frac{1}{2y} \left(\frac{T}{A} \right)^2$$

$$U = \frac{T^2}{2Ay} \cdot \ell$$

$$U_1 = U_2$$

$$\Rightarrow \frac{1}{2} \epsilon_1^2 y_1 \times A \ell = \frac{1}{2} \epsilon_2^2 y_2 \times A \times 2\ell$$

$$\frac{y_2}{y_1} = \frac{1}{2} \left(\frac{\epsilon_1}{\epsilon_2} \right)^2 = 8$$

$$(S) u_1 = u_2$$

$$\frac{1}{2} \times \epsilon_1^2 \times y_1 = \frac{1}{2} \times \epsilon_2^2 \times y_2$$

$$\frac{y_2}{y_1} = \left(\frac{\epsilon_1}{\epsilon_2} \right)^2 = 16$$

11. **Ans (C)**

In KL, V = constant, W = 0, Q > 0

In LM, T = constant, W > 0, Q > 0

In ML, P = constant, W < 0, Q < 0

In JK, T = constant, W < 0, Q < 0

PART-1 : PHYSICS

SECTION-II

1. Ans (594)

$$\text{Least count} = \frac{P}{N} = \frac{1\text{mm}}{100} = 0.01 \text{ mm}$$

Diameter of the wire is

$$d = 4\text{mm} + 56 \times \text{L.C} - 6 \times \text{L.C}$$

$$d = 4\text{mm} + 50 \times \text{L.C}$$

$$d = 4\text{mm} + 50 \times 0.01 \text{ mm}$$

$$d = 4.50\text{mm}$$

$$d = 0.450 \text{ cm}$$

The curved surface area of the wire is

$$S = \pi dl$$

$$S = \frac{22}{7} \times 0.450 \times 4.2 = 5.94 \text{ cm}^2$$

2. Ans (4)

The tower will be stable if its center of mass is located within its area of support. If we set the origin of our coordinate system to be the bottom left corner, that's equivalent to saying

$$x_{\text{cm}} \leq b$$

Calculating x_{cm} :

$$x_{\text{cm}} = \frac{\frac{b}{2} + \frac{b}{2} + \frac{a}{2}}{2} = \frac{2b + a}{6}$$

Yields :

$$2b + a \leq 6b$$

$$\frac{a}{b} \leq 4$$

3. Ans (3)

$$\sqrt{2g(1-x)h} \times \sqrt{\frac{2xh}{g}} + \sqrt{2g(1-x)h} \times \sqrt{\frac{2(1+x)h}{g}}$$

$$= 2h\sqrt{x(1-x)} + 2h\sqrt{1-x^2}$$

$$y = \sqrt{x(1-x)} + \sqrt{1-x^2}$$

$$\frac{dy}{dx} = 0 = \frac{1-2x}{2\sqrt{x(1-x)}} + \frac{-2x}{2\sqrt{1-x^2}}$$

$$\frac{1-2x}{\sqrt{x(1-x)}} = \frac{2x}{\sqrt{1-x^2}}$$

$$\frac{(1-2x)^2}{x} = \frac{4x^2}{1+x}$$

$$4x^3 = (1+x)(1+4x^2-4x)$$

$$0 = -3x + 1 \Rightarrow x = \frac{1}{3}$$

4. Ans (1)

The magnetic field from the wire is given by

$$B = \frac{\mu_0 i_1}{2\pi x}$$

Let θ be the direction of a component of force from the vertical. It is then seen that

$$dF = Bi_2 d\ell \Rightarrow dF_x = Bi_2 d\ell \sin\theta = Bi_2 dy$$

We only consider the force in the x-direction

which means that

$$F_x = \int_{-\infty}^{\infty} dF_x = \frac{\mu_0 i_1 i_2}{2\pi} \int_{-\infty}^{\infty} \frac{dy}{x}$$

Solving the equation in terms of x and then

plugging in gives us

$$F_x = \frac{8\mu_0 i_1 i_2}{2\pi} \int_{-\infty}^{\infty} \frac{dy}{y^2 - 6y + 25}$$

$$= \frac{8\mu_0 i_1 i_2}{2\pi} \cdot \frac{\pi}{4} = \mu_0 i_1 i_2$$

5. Ans (1)

$$m_1 m_2 = \frac{v_1}{-u} \times \frac{-(u+d)}{(v_1-d)} = -1$$

$$1 + \frac{d}{u} = \frac{d}{v_1} - 1$$

$$2 + \frac{d}{u} = \frac{d}{v_1}$$

$$v_1 = \frac{ud}{d+2u} \quad \dots (1)$$

$$v_1 - d = \frac{-ud - d^2}{d+2u} = \frac{-d(d+u)}{(d+2u)} \quad \dots (2)$$

$$\frac{1}{f_1} = \frac{d+2u}{ud} + \frac{1}{u} = 2 \left(\frac{1}{d} + \frac{1}{u} \right) = \frac{2(d+u)}{du} \quad \dots (3)$$

$$\frac{1}{f_2} = \frac{-1}{u+d} + \frac{d+2u}{d(d+u)}$$

$$\frac{1}{f_2} = \frac{-d+d+2u}{d(d+u)} = \frac{2u}{d(d+u)} \quad \dots (4)$$

$$(3) \times (4) \Rightarrow \frac{1}{f_1 f_2} = \frac{4}{d^2}$$

$$d = 2\sqrt{f_1 f_2}$$

$$\frac{1}{f_1} = \frac{2(d+u)}{du}$$

$$du = 2f_1 d + 2uf_1$$

$$u = \frac{2f_1 d}{d-2f_1} = \frac{2f_1 \times 2\sqrt{f_1 f_2}}{2\sqrt{f_1 f_2} - 2f_1}$$

$$= \frac{2f_1 \sqrt{f_2}}{\sqrt{f_2} - \sqrt{f_1}}$$

$$d = 2\sqrt{f_a f_b}, u = \frac{2f_a \sqrt{f_b}}{\sqrt{f_b} - \sqrt{f_a}}$$

$$\frac{u}{d} = 1$$

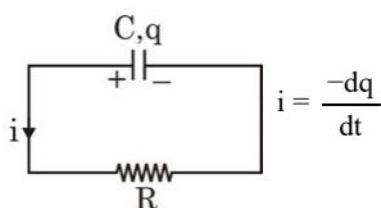
6. Ans (22)

$$10 = 6A + B$$

$$4 = 0.06A + B$$

$$\Rightarrow B \approx 4, A \approx 1$$

$$\therefore R = V + 4$$



$$iR = \frac{q}{C} = V$$

$$\Rightarrow \frac{-dq}{dt} \left(\frac{q}{C} + 4 \right) = \frac{q}{C}$$

$$\Rightarrow \int_{q_0}^{q_0/100} -dq \cdot \frac{(q+4)}{q} = \int_0^t dt \quad (q_0 = 6C)$$

$$\Rightarrow t = - \left[\frac{q_0}{100} - q_0 + 4 \ln \left(\frac{1}{100} \right) \right]$$

$$= \frac{99q_0}{100} + 4 \ln 100 = \frac{99 \times 6}{100} + 8 \times 2$$

$$= 16 + 5.94 = 21.94 \approx 22 \text{ s}$$

PART-2 : CHEMISTRY

SECTION-I (i)

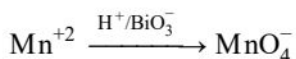
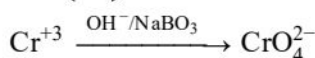
1. Ans (A)

$$\Lambda_m \text{ of } \text{HCl} > \text{KCl} > \text{LiCl} > \text{CH}_3\text{COOH}$$

2. Ans (B)

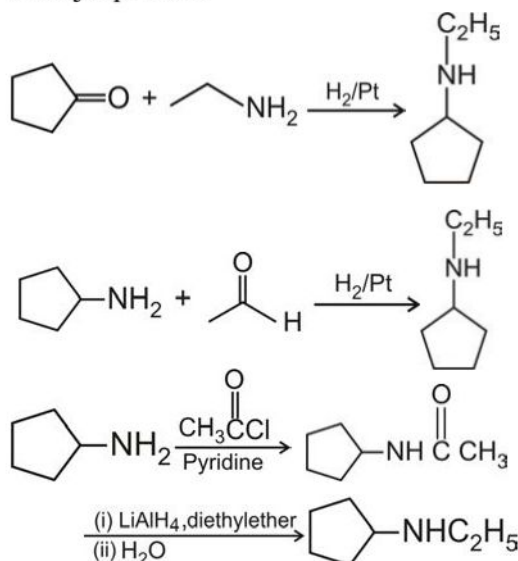
Kuchero reaction

3. Ans (D)



4. **Ans (D)**

Reaction (a, b, c) all give N-ethylcyclopentylamine as major product.



PART-2 : CHEMISTRY

SECTION-I (ii)

5. **Ans (B,C)**

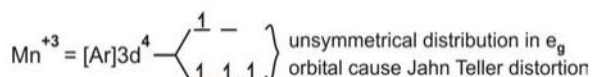
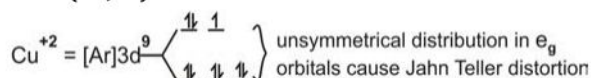
$$P_{\text{total}} = \frac{P_A}{Y_A} = \frac{P_B}{Y_B}$$

$$\frac{X_A P_A^{\circ}}{Y_A} = \frac{(1 - X_A) P_B^{\circ}}{(1 - Y_A)}$$

$$\frac{1}{X_A} = \frac{1}{Y_A} \times \frac{P_A^{\circ}}{P_B^{\circ}} + \frac{P_B^{\circ} - P_A^{\circ}}{P_B^{\circ}}$$

$$\text{Slope} = \frac{P_A^{\circ}}{P_B^{\circ}}, \text{Intercept} = \frac{P_B^{\circ} - P_A^{\circ}}{P_B^{\circ}}$$

7. **Ans (A,C)**



PART-2 : CHEMISTRY

SECTION-I (iii)

8. **Ans (C)**

Intermolecular force of attraction $\propto a$

Size of molecule $\propto b$

$$\text{Boyle's temperature } (T_B) = \frac{a}{Rb}$$

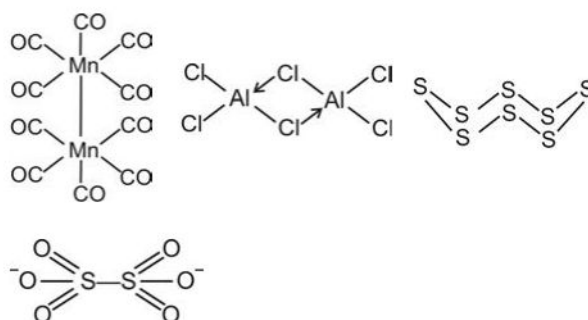
$$\text{Critical temperature } (T_C) = 8 \times T_B$$

9. **Ans (A)**

Strongly basic nucleophiles (II, III) cannot be used in presence of protic solvents (P,T)

Neutral Nucleophiles (IV) work better in presence of less protic solvents or aprotic solvents

10. **Ans (B)**



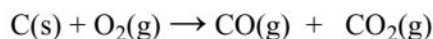
PART-2 : CHEMISTRY

SECTION-II

1. **Ans (4)**

1, 3, 5, 7 statement are correct.

2. **Ans (160)**

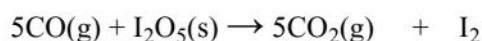


a mole b mole

$$\text{POAC on carbon, } 1 \times 0.1 = a + b$$

$$\text{POAC on oxygen, } 2 \times 0.08 = a + 2b$$

$$a = 0.04 \text{ \& } b = 0.06$$



0.04/5 mole

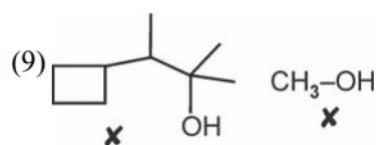
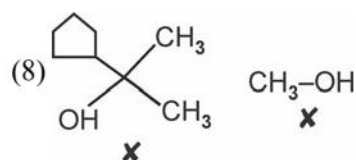
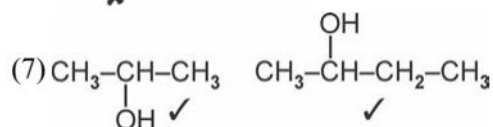
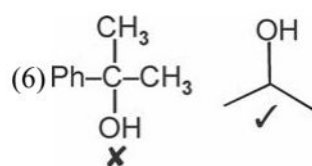
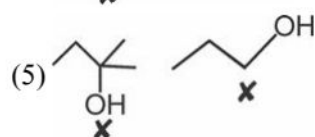
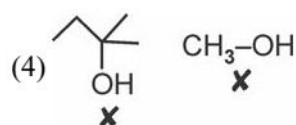
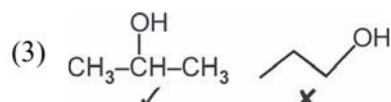
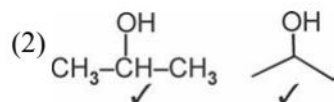
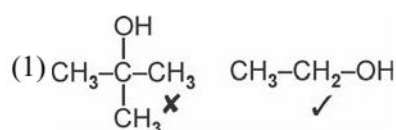
gmequivalent of I_2 = gmequivalent of hypo

$$0.008 \times 2 = 0.1 \times x \times 10^{-3}$$

$$x = 160 \text{ ml}$$

3. Ans (3)

(I, III, VI)



4. Ans (331)

X = KBr

P = HBr

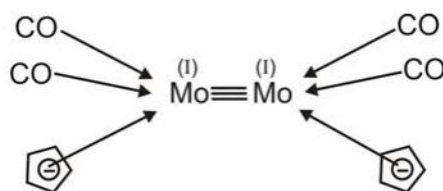
Q = Br₂

S = Pb(CH₃COO)₂

R = PbBr₂

Y = 2,4,6 tribromophenol

5. Ans (3)

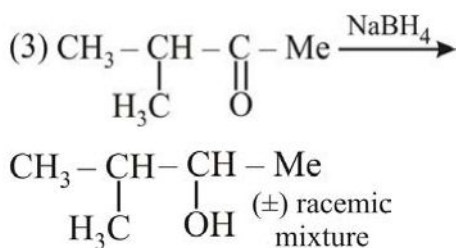
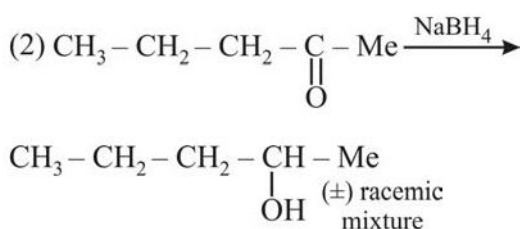
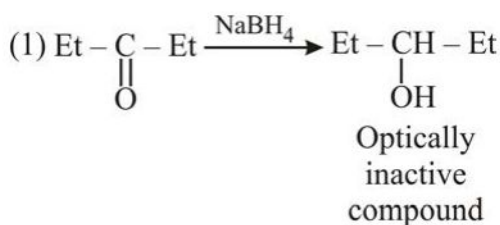


In above strength both Mo have 18 valence electrons

6. Ans (2)

M. wt 86 of ketone

So m. formula = C₅H₁₀O



(2 ; 3)

PART-3 : MATHEMATICS

SECTION-I (i)

1. **Ans (B)**

$$\text{Let } I = \int_{\frac{1}{4}}^4 \frac{\tan^{-1}x}{x} dx \quad \dots(1)$$

$$\text{put } x = \frac{1}{t} \Rightarrow dx = -\frac{1}{t^2} dt$$

$$\Rightarrow I = \int_{\frac{1}{4}}^4 \frac{\cot^{-1}x}{x} dx \quad \dots(2)$$

$$\therefore (1) + (2) \Rightarrow 2I = \frac{\pi}{2} \int_{\frac{1}{4}}^4 \frac{dx}{x}$$

$$\Rightarrow I = \frac{\pi}{4} \left(\ln 4 - \ln \frac{1}{4} \right) \Rightarrow I = \frac{\pi}{2} \ln 4$$

$$= \pi \ln 2$$

$$= 3.14 \times 0.69$$

$$= 2.166$$

$$\therefore 2[I] = 4$$

2. **Ans (C)**

$$\text{Put } x^2 = t \Rightarrow \int_0^1 f(t) dt = 1$$

$$\therefore \int_0^1 f(x) dx + \int_{f(0)}^{f(1)} f^{-1}(x) dx = 1.f(1) - 0.f(0)$$

$$\Rightarrow 1 + \int_1^2 f^{-1}(x) dx = 2$$

$$\Rightarrow \int_1^2 f^{-1}(x) dx = 1$$

3. **Ans (C)**

$$\text{Let } \tan \theta = t \Rightarrow \theta \in [0, \pi/2) \cup \{\pi, 2\pi\}$$

$$\Rightarrow \sqrt{t} = 2 \sin \theta$$

squaring

$$\left(1 + \frac{1}{t^2}\right)t = 4$$

$$t + \frac{1}{t} = 4 \Rightarrow t = 2 + \sqrt{3}, 2 - \sqrt{3}$$

$$\Rightarrow \theta = 0, \frac{\pi}{12}, \frac{5\pi}{12}, \pi, 2\pi$$

4. **Ans (C)**

$$I = \lim_{n \rightarrow \infty} \int_0^{\pi/2} n \left(1 - (\sin x)^{\frac{1}{n}}\right) dx$$

$$= \int_0^{\pi/2} \left(\lim_{k \rightarrow 0} \frac{1 - (\sin x)^k}{k} \right) dx$$

$$= \int_0^{\pi/2} \left(\lim_{k \rightarrow 0} \frac{-(\sin x)^k \ln(\sin x)}{1} \right) dx$$

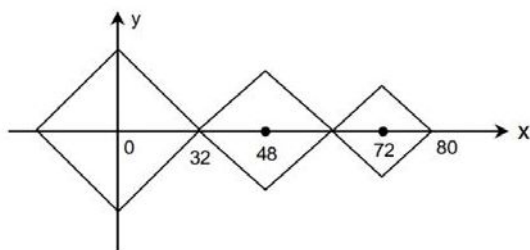
$$= - \int_0^{\pi/2} \ln(\sin x) dx$$

$$= \frac{\pi}{2} \ln 2$$

PART-3 : MATHEMATICS

SECTION-I (ii)

5. Ans (A,C)



$$a_2 = a_1 + \frac{3}{2} b_1 = 48, b_2 = \frac{b_1}{2} = 16$$

$$a_3 = 48 + \frac{3}{2} \times 16 = 72, b_3 = \frac{16}{2} = 8$$

So, the three loops from $i = 1$ to $i = 3$ are alike

Now, area of i^{th} loop (square) = $\frac{1}{2} (\text{diagonal})^2$

$$A_i = \frac{1}{2} (2b_i)^2 = 2b_i^2$$

So, the areas form a G.P series

$\therefore$ sum of the G.P up to infinite term is

$$\frac{A_i}{1-r} = 2(32)^2 \times 1 \frac{1}{1-\frac{1}{4}} = \frac{8}{3}(32)^2 \text{ sq. units}$$

6. Ans (A,B,C,D)

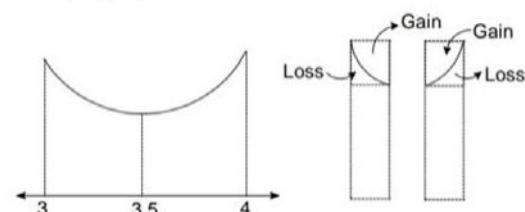
$$\because y = mx - 2m - m^3$$

$$\Rightarrow 6 = 9m - 2m - m^3$$

$$\Rightarrow m^3 - 7m + 6 = 0 \Rightarrow m = 1, +2, -3$$

$$\therefore \text{Centre is } \left(\frac{11}{2}, \frac{3}{2} \right)$$

7. Ans (A,C)



$$f(x) = x^2 - 7x + 13$$

In both cases gain > loss

Curve is symmetrical about line $x = \frac{7}{2}$

$$\Rightarrow P > \int_3^4 f(x) dx \text{ \& } Q > \int_3^4 f(x) dx$$

PART-3 : MATHEMATICS

SECTION-I (iii)

8. Ans (D)

(P) Given $a, d \in \mathbb{I}^+$ and $a + 2d = 8 \Rightarrow a = 8 - 2d$

$$\text{also, } 191 < \frac{15}{2} (2a + 14d) < 196$$

$$\Rightarrow \frac{191}{15} < a + 7d < \frac{196}{15}$$

$$\Rightarrow \frac{191}{15} < 8 + 5d < \frac{196}{15} \Rightarrow \frac{71}{75} < d < \frac{76}{75} \because d \in \mathbb{I}^+$$

$\therefore d = 1$ and $a = 6$

(Q) $(2\cos^2 y - 1) - (2\cos^2 x - 1) = 3a - 2$

$$\cos^2 y - \cos^2 x = \frac{3a - 2}{2} \quad \dots(1)$$

$$\cos^2 x + \sin^2 y = \frac{a^2}{2} \quad \dots(2)$$

equation (1) + (2)

$$1 = \frac{a^2 + 3a - 2}{2} \Rightarrow a^2 + 3a - 4 = 0$$

$a = 1$ & $a = -4$ (impossible)

$$(R) \sin^{-1} \frac{2x}{1+x^2} = \frac{\pi}{2} + \cos^{-1} x$$

solution will exist if $\cos^{-1} x = 0$

$$\& \sin^{-1} \frac{2x}{1+x^2} = \frac{\pi}{2}$$

which is possible only when $x = 1$.

$$(S) \frac{1}{\Delta} \sqrt{abc(a+b+c)} = \frac{1}{\Delta} \sqrt{4R \cdot \Delta \cdot 2S}$$

$$= \sqrt{\frac{8RS}{\Delta}} = \sqrt{\frac{8R}{r}} \geq \sqrt{8 \cdot 2} = 4$$

$$\because r = 4R \sin \frac{A}{2} \sin \frac{B}{2} \sin \frac{C}{2}$$

$$\frac{r}{R} = 4 \sin \frac{A}{2} \sin \frac{B}{2} \sin \frac{C}{2} \Rightarrow \frac{R}{r} \geq 2.$$

$$\text{In any triangle } \sin \frac{A}{2} \sin \frac{B}{2} \sin \frac{C}{2} \leq \frac{1}{8}$$

9. Ans (B)

$$(P) \int_{-2}^6 f(\{x\}) dx = \int_{-2}^2 f(\{x\}) dx + \int_2^6 f(\{x\}) dx$$

$$\therefore \int_2^6 f(\{x\}) dx = -2$$

$$(Q) |\vec{a} + \vec{b}|^2 + |\vec{a} - \vec{b}|^2 = 2(|\vec{a}|^2 + |\vec{b}|^2)$$

$$\Rightarrow |\vec{a}| = 5$$

$$(R) 2 \int_0^{\pi} |\operatorname{sgn}(\tan x) + \operatorname{sgn}(\cot x)| dx$$

$$2 \left(\int_0^{\pi/2} 2 dx + \int_{\pi/2}^{\pi} 2 dx \right) = 8 \cdot \frac{\pi}{2} = 4\pi$$

$$(S) A = \begin{bmatrix} 1 & \left(\frac{1}{2} + \frac{1}{2^2} + \dots \right) \\ 0 & 1 \end{bmatrix}$$

Sum of all elements of matrix A = 3

10. Ans (A)

$$(P) \text{ Required probability} = \frac{{}^{n-3}C_4}{{}^nC_4}$$

$$(Q) P_2 = \frac{n-3}{n} \times \frac{3}{(n-1)} \text{ or}$$

$$\left(\frac{{}^{n-3}C_1}{{}^nC_1} \cdot \frac{{}^3C_1}{{}^{n-1}C_1} \right)$$

$$(R) P_{n-2} = \frac{{}^{n-3}C_{n-3}}{{}^nC_{n-3}} = \frac{1}{{}^nC_3}$$

$$(S) P_r = \frac{{}^{n-3}C_{r-1}}{{}^nC_{r-1}} \cdot \frac{3}{(n+1-r)} \text{ or}$$

$$\left(P_r = \frac{{}^{n-3}C_{r-1}}{{}^nC_{r-1}} \cdot \frac{{}^3C_1}{{}^{n-r+1}C_1} \right)$$

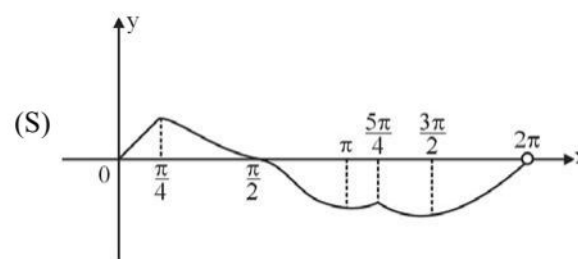
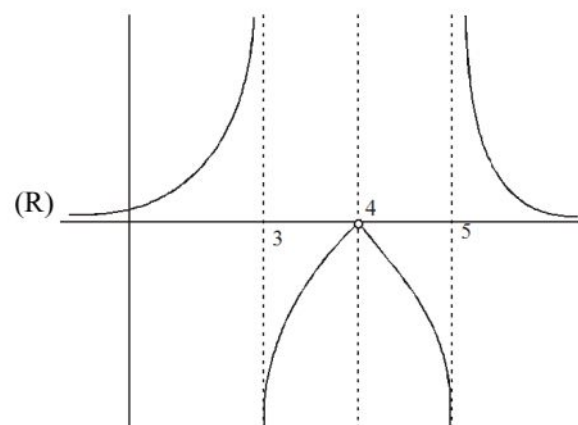
11. Ans (B)

$$(P) \lim_{x \rightarrow 0} \left[\min. \left((y+1)^2 + 3 \right) \cdot \frac{\sin x}{x} \right]$$

$$= 2$$

$$(Q) \lim_{x \rightarrow 0} \frac{\sin^{-1} \left(\frac{2x}{1+x^2} \right)}{\left(\frac{2x}{1+x^2} \right)} \cdot \frac{2x}{x \cdot (1+x^2)}$$

$$= 1 \cdot \frac{2}{1+0} = 2$$



PART-3 : MATHEMATICS

SECTION-II

1. Ans (75)

$$1 + {}^4C_1 \times 2 + \frac{|4|}{|2|} \times \frac{|4|}{|2|} \times \frac{|3|}{|2|} + \frac{|4| \times |3|}{|2| \times |1| \times |1| \times |2|} + |4| = 75$$

or

$$1 + {}^4C_3 |2| + {}^4C_2 |3| + |4|$$

2. Ans (3)

By properties $PA = PB = \sqrt{10}$

$$\frac{x-1}{-1/\sqrt{10}} = \frac{y-5}{3/\sqrt{10}} = \pm\sqrt{10}$$

$$\Rightarrow A \equiv (0, 8); B \equiv (2, 2)$$

Slopes of asymptotes are -7 , and 1

If angle between asymptotes be θ then

$$\tan \theta = \left| \frac{-7-1}{1-7} \right| = \frac{4}{3}$$

$$\text{Now, } 2\tan^{-1} \frac{b}{a} = \tan^{-1} \frac{4}{3} \Rightarrow \frac{\frac{2b}{a}}{1 - \frac{b^2}{a^2}} = \frac{4}{3}$$

$$\Rightarrow \frac{b}{a} = \frac{1}{2} \quad \left(\frac{b}{a} = 2 \text{ rejected} \right) \quad \dots(1)$$

$$\frac{b^2}{a^2} = e^2 - 1 \Rightarrow e = \frac{\sqrt{5}}{2}$$

Also, area of $\triangle CAB = ab$

$$ab = \left| \frac{1}{2} [1(8-2) + 0(2-1) + 2(1-8)] \right| = 4 \quad \dots(2)$$

$$b^2 = 2, a = 2b = 2\sqrt{2} \text{ length of latus rectum}$$

$$= \frac{2b^2}{a} = \frac{4}{2\sqrt{2}} = \sqrt{2} = \ell$$

$$\therefore [e^2 + \ell^2] = 3$$

3. Ans (1)

point of intersection of lines at origin $O = (0, 0, 0)$,

$$A = (3, 2, 2) \text{ \& } B = \left(\frac{8}{3}, 2, \frac{4}{3} \right)$$

$$\text{Area of triangle} = \frac{1}{2} \left| \overrightarrow{OA} \times \overrightarrow{OB} \right|$$

$$\text{Area of triangle} = \frac{1}{2} \left| -\frac{4}{3}\hat{i} + \frac{4}{3}\hat{j} + \frac{2}{3}\hat{k} \right| = 1$$

4. Ans (6)

$$A^{-1}B^2 + AB = 0$$

$$\Rightarrow A^{-1}B.BB^{-1} + AB.B^{-1} = 0$$

$$\Rightarrow AA^{-1}B + A.A = 0$$

$$\Rightarrow B = -A^2 \quad \dots\dots(1)$$

$$\& |A| = 5 \quad \dots\dots(2)$$

$$A^6 - 2A^4B + A^2B^2 = 4A^6$$

$$\therefore |4A^6| = 4^3 |A|^6 = 2^6 \cdot 5^6 = 10^6$$

$$\therefore n = 6$$

5. Ans (1)

$$(z+1) \begin{vmatrix} z+\omega^2 & 1 \\ 1 & z+\omega \end{vmatrix} + \omega \begin{vmatrix} 1 & \omega \\ z+\omega & \omega^2 \end{vmatrix} + \omega^2 \begin{vmatrix} \omega & z+\omega^2 \\ \omega^2 & 1 \end{vmatrix} = 0$$

$$\text{As } 1 + \omega + \omega^2 = 0, \quad \omega^4 = \omega$$

$$\Rightarrow (z+1)(z^2 + z(\omega + \omega^2) + \omega^3 - 1) +$$

$$\omega(\omega^2 - z\omega - \omega^2) + \omega^2(\omega - z\omega^2 - \omega^4) = 0$$

$$\Rightarrow (z+1)(z^2 - z) + z(-\omega^2 - \omega) = 0$$

$$\Rightarrow z^3 - z^2 + z^2 - z + z = 0$$

$$\Rightarrow z^3 = 0$$

$$\Rightarrow z = 0 \text{ is only 1 complex number}$$

6. Ans (0)

Replacing x by $1+x$ in (1), we get

$$f'(1+x) = -f'(1-x)$$

$$\text{Let } I = \int_{-1}^1 f'(1+x)x^2e^{x^2}dx \quad \dots\dots(2)$$

$$I = \int_{-1}^1 f'(1-x).x^2e^{x^2}dx \text{ (using king)}$$

$$I = - \int_{-1}^1 f'(1+x).x^2e^{x^2}dx$$

$$(\because f'(1+x) = -f'(1-x))$$



CLASSROOM CONTACT PROGRAMME

(Academic Session : 2023 - 2024)

JEE (Advanced)

AIOT

21-04-2024

JEE(Main + Advanced) : ENTHUSIAST & LEADER COURSE (SCORE ADVANCED)

ANSWER KEY

PAPER-2 (OPTIONAL)

PART-1 : PHYSICS

SECTION-I (i)	Q.	1	2	3	4		
	A.	B	C	B	C		
SECTION-I (ii)	Q.	5	6	7			
	A.	A,C,D	A,B	B,C			
SECTION-II (i)	Q.	1	2	3	4		
	A.	3.00	3.12 to 3.16	0.80	2.80 to 2.81		
SECTION-II (ii)	Q.	5	6	7	8	9	10
	A.	3	16	9	3	8	114

PART-2 : CHEMISTRY

SECTION-I (i)	Q.	1	2	3	4		
	A.	B	D	A	C		
SECTION-I (ii)	Q.	5	6	7			
	A.	B,C	B,C,D	A,B,C,D			
SECTION-II (i)	Q.	1	2	3	4		
	A.	2.60	42.50	8.00	7.10		
SECTION-II (ii)	Q.	5	6	7	8	9	10
	A.	300	5	5	3	7	2

PART-3 : MATHEMATICS

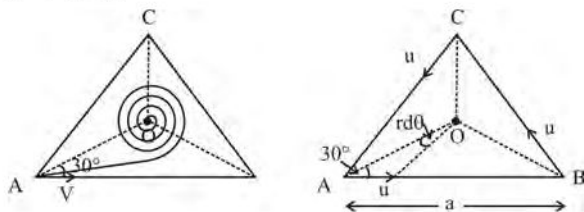
SECTION-I (i)	Q.	1	2	3	4		
	A.	C	D	A	B		
SECTION-I (ii)	Q.	5	6	7			
	A.	A,B,C,D	B,D	B,D			
SECTION-II (i)	Q.	1	2	3	4		
	A.	2.00	9.00	67.00	22.00		
SECTION-II (ii)	Q.	5	6	7	8	9	10
	A.	3	0	80	4	4	5

HINT – SHEET

PART-1 : PHYSICS

SECTION-I (i)

1. Ans (B)



$$\frac{dr}{dt} = -v \cos 30^\circ = -\frac{\sqrt{3}}{2} v$$

$$r \frac{d\theta}{dt} = v \sin 30^\circ = \frac{v}{2}$$

$$\frac{1}{r} \frac{d\theta}{dr} = -\sqrt{3}$$

$$\int_{r_0}^r \frac{dr}{r} = -\sqrt{3} \int_{\theta_0}^{\theta} d\theta \Rightarrow r = r_0 e^{-\sqrt{3}\theta}$$

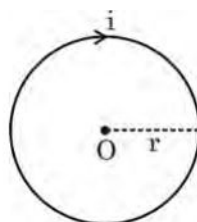
When A completes one revolution $\theta = 2\pi$

$$\text{Time taken } t = \frac{r_0 (1 - e^{-2\sqrt{3}\pi})}{\sqrt{3}v/2}$$

$$\text{Distance travelled } D = vt = \frac{2r_0}{\sqrt{3}} (1 - e^{-2\sqrt{3}\pi})$$

$$D = \frac{2a}{3} (1 - e^{-2\sqrt{3}\pi})$$

2. Ans (C)



$$B_0 = \frac{\mu_0 i}{2\pi r}$$

$$\phi = B_0 \pi r^2$$

$$\epsilon = \frac{\Delta\phi}{\Delta t} = \frac{\mu_0}{2\pi r} \pi r^2 \frac{\Delta i}{\Delta t} = \frac{\mu_0 r}{2} \frac{\Delta i}{\Delta t}$$

$$\text{Res.} = \frac{\epsilon}{i} \Rightarrow \frac{\rho \ell}{A} = \frac{\epsilon}{i} \Rightarrow \rho = \frac{\epsilon A}{\ell i}$$

$$\rho = \frac{\mu_0 r}{2} \frac{\Delta i}{\Delta t} \times \frac{A}{2\pi r} \times \frac{1}{i}$$

$$\rho = \frac{\mu_0}{4\pi} \times \frac{\Delta i}{\Delta t} \times A \times \frac{1}{i}$$

$$r = 5 \text{ cm}$$

$$A = \frac{\pi d^2}{4}, d = 1 \text{ mm}$$

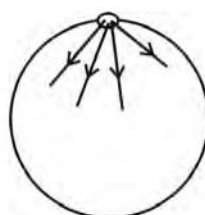
$$\frac{\Delta i}{\Delta t} = \frac{10^{-6}}{365 \times 86400} \frac{A}{s}$$

$$i = 100 \text{ A}$$

$$\rho = \frac{10^{-7} \times 3.17 \times 10^{-14} \times 0.78 \times 10^{-6}}{10^2}$$

$$\rho = 2.5 \times 10^{-29}$$

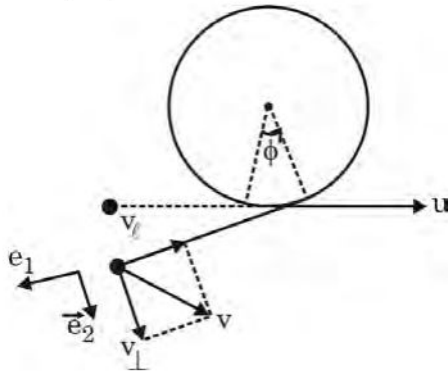
3. Ans (B)



$$\text{Power received} = I_0 \pi r^2$$

$$\text{Intensity} = \frac{I_0 \pi r^2}{4\pi R^2} = \frac{I_0 r^2}{4R^2}$$

4. Ans (C)



$\vec{e}_1$ and $\vec{e}_2$ are unit vectors along and perpendicular to the thread.

$$\vec{v} = -v_{\parallel}\vec{e}_1 + v_{\perp}\vec{e}_2 \dots\dots(1)$$

$$\vec{a} = \frac{d\vec{v}}{dt} = -u\frac{d\vec{e}_1}{dt} - \underbrace{\frac{du}{dt}\vec{e}_1}_0 + v_{\perp}\frac{d\vec{e}_2}{dt} + \frac{dv_{\perp}}{dt}\vec{e}_2$$

$$\vec{a} = -u\frac{d\vec{e}_1}{dt} + v_{\perp}\frac{d\vec{e}_2}{dt} + \frac{dv_{\perp}}{dt}\vec{e}_2$$

vector $\vec{e}_1$ and $\vec{e}_2$ rotates as a rigid body with angular velocity

$$\vec{\omega} = \frac{d\phi}{dt}(\vec{e}_1 \times \vec{e}_2)$$

$$\frac{d\vec{e}_1}{dt} = \vec{\omega} \times \vec{e}_1 = \frac{d\phi}{dt}\vec{e}_2$$

$$\frac{d\vec{e}_2}{dt} = \vec{\omega} \times \vec{e}_2 = -\frac{d\phi}{dt}\vec{e}_1$$

$$\vec{a} = -v_{\perp}\frac{d\phi}{dt}\vec{e}_1 + \left(-u\frac{d\phi}{dt} + \frac{dv_{\perp}}{dt}\right)\vec{e}_2$$

now there is no force in direction $\vec{e}_2$

$$\Rightarrow \frac{dv_{\perp}}{dt} = u\frac{d\phi}{dt} \Rightarrow v_{\perp} = u\phi$$

When P is about to detach

$$\phi = 2N\pi$$

$$v_{\perp} = 2N\pi u$$

$$v = \sqrt{v_{\parallel}^2 + v_{\perp}^2} = u\sqrt{1 + (2N\pi)^2}$$

PART-1 : PHYSICS

SECTION-I (ii)

5. Ans (A,C,D)

$$\text{Frequency of surface MN}(f') = \frac{10f}{7}$$

$$\lambda_A = \frac{7v}{f'} = \frac{49v}{10f}$$

$$\lambda_B = \frac{7v}{f'} = \frac{49v}{10f}$$

$$f_A = \left(\frac{7v - \frac{v}{6}}{7v}\right)f' = \left(\frac{205}{147}\right)f$$

$$f_B = \left(\frac{7v + \frac{v}{9}}{7v}\right)f' = \left(\frac{640}{441}\right)f$$

6. Ans (A,B)

Let ω_0 be the angular velocity of the rod just before the collision. Then from conservation of mechanical energy:

$$\frac{1}{2} \frac{ML^2}{3} \omega_0^2 = Mg \frac{L}{2} (1 - \sin \theta)$$

$$\text{When } \sin \theta = \frac{L}{2L} = \frac{1}{2}$$

$$\Rightarrow \omega_0 = \sqrt{\frac{3g}{2L}}$$

because of hinge reaction, linear momentum of the system cannot be conserved in the horizontal direction.

But angular momentum of the system can be conserved about the hinge, just during the collision, neglecting the effect of gravity and its torque about the hinge during the collision time because gravity is non-impulsive force.

If ω = angular velocity of the (Rod + Bullet) system just after collision,

$$\text{Then } \left(\frac{ML^2}{3} + mL^2 \right) \omega = \frac{ML^2}{3} \omega_0 + Mv_0 \frac{L}{2}$$

$$\frac{1}{2} \left(\frac{ML^2}{3} + mL^2 \right) \omega^2 = Mg \frac{L}{4} + mg \frac{L}{2}$$

$$\text{putting } M = 3m : \omega^2 L = \frac{5g}{4} \dots (iii)$$

using (i), (ii) and (iii)

$$v_0 = \sqrt{gL} (\sqrt{6} + \sqrt{20})$$

(b) After collision, when the system makes an

angle 60° with the horizontal, total torque about

$$\text{the hinge} = Mg = \frac{L}{2}$$

$$\cos 60^\circ + mg L \cos 60^\circ = \frac{5}{4} mgL,$$

by putting $M = 3m$

Moment of inertia of the system about the

$$\text{hinge} = \frac{ML^2}{3} + mL^2 = 2mL^2$$

By putting $M = 3m$

Hence, angular acceleration of the system

$$= \frac{\text{Torque}}{\text{Moment of inertia}} = \frac{(5/4)mgL}{2mL^2} = \frac{5g}{8L}$$

Therefore, tangential acceleration of the

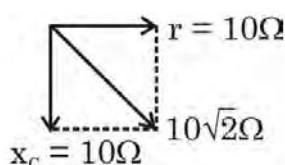
midpoint

$$= \left(\frac{L}{2} \right) \left(\frac{5g}{8L} \right) = \frac{5g}{16}$$

7. Ans (B,C)

(1) In the source

$$x_C = \frac{1}{\omega C} = 10\Omega, r = 10\Omega$$



$$z_1 = r - x_C j$$

(2) Let load has $z_2 = R + xj$

$$z \text{ of existing ckt} \Rightarrow z_1 = r + (-x_C)j$$

$$z_{eq} = z_1 + z_2 \Rightarrow z_{eq} = (R + r) + (x - x_C)j$$

$$P_{load} = i_{rms \text{ load}} v_{rms \text{ load}} \cos \phi_{load}$$

$$P_{load} = \frac{\epsilon_{rms}^2}{z_{eq}^2} R$$

$$P_{load} = \frac{\epsilon_{rms}^2 R}{\sqrt{(R+r)^2 + (x-x_C)^2}}$$

for P_{load} to be maximum

$$x = x_C = 10\Omega$$

$$P_{load} = \frac{\epsilon_{rms}^2 R}{(R+r)^2}$$

$$\frac{dP_{load}}{dR} = 0$$

$$\Rightarrow R = r$$

$$\Rightarrow (P_{load})_{max} = \frac{\epsilon_{rms}^2}{4R}$$

$$\text{Now } R = r = 10\Omega$$

$$\epsilon_{rms} = \frac{15}{\sqrt{2}} \text{ volt}$$

$$(P_{load})_{max} = \frac{15 \times 15}{2 \times 4 \times 10}$$

$$(P_{load})_{max} = \frac{45}{16} \text{ watt}$$

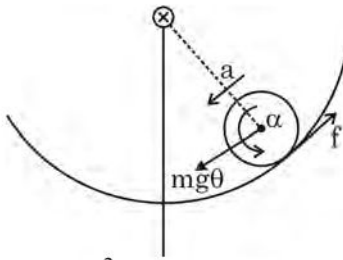
$$z_2 = z_{load} = 10 + 10j$$

$$P.f. = \phi = \frac{10}{10\sqrt{2}} = \frac{1}{\sqrt{2}}$$

PART-1 : PHYSICS

SECTION-II (i)

2. Ans (3.12 to 3.16)

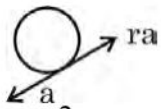


$$fr = \frac{mr^2}{2}\alpha \rightarrow f = \frac{mr\alpha}{2} \dots\dots(1)$$

$$mg\theta - f = ma$$

$$mg\theta - \frac{ma}{2} = ma$$

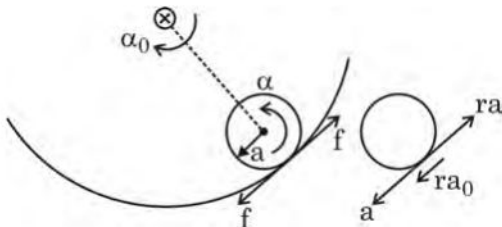
$$mg\theta = \frac{3}{2}ma$$



$$a = \frac{2g}{3}\theta \rightarrow \theta = \frac{2g}{3(R-r)}\theta$$

$$\omega = \sqrt{\frac{2g}{3(R-r)}} = \sqrt{\frac{2 \times 1000}{3 \times 0.75}} = \sqrt{\frac{16 \times 5}{9}}$$

$$= \frac{4}{3} \times 2.25 = \frac{4}{3} \times \frac{225}{100} = 3$$



$$a - r\alpha = R\alpha_0 \dots\dots(iii)$$

$$fr = \frac{mr^2}{2}\alpha \rightarrow f = \frac{mr\alpha}{2}$$

$$mg\theta - f = ma \dots\dots(4)$$

$$fR = MR^2\alpha_0$$

$$f = MR\alpha_0$$

Put from (iii)

$$f = M(a - r\alpha) \dots\dots(iv)$$

by (i) & (iv)

$$\frac{mr\alpha}{2} = Ma - Mr\alpha$$

$$\left(\frac{m}{2} + M\right)r\alpha = Ma$$

$$\gamma\alpha = \frac{M}{\left(\frac{m}{2} + M\right)}a$$

Put in equation (ii)

$$mg\theta - \frac{m}{2}\left(\frac{2m}{m+2M}\right)a = ma$$

$$g\theta = a + \left(\frac{M}{m+2M}\right)a$$

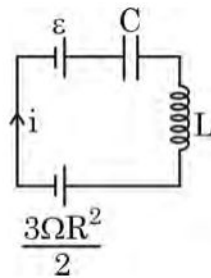
$$\theta = \frac{g(m+2M)}{(m+3m)(R-r)}\theta$$

$$\omega = \sqrt{\left(\frac{m+2M}{m+3M}\right)\frac{g}{(R-r)}}$$

Given $M = 1\text{kg}$, $m = 1\text{kg}$

$$\omega = \sqrt{10}$$

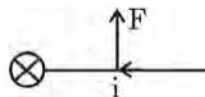
3. Ans (0.80)



$$\epsilon - \frac{q}{C} - L\frac{di}{dt} - \frac{3R^2}{2}\Omega = 0$$

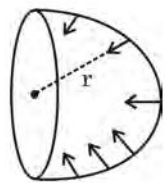
$$\frac{d\epsilon}{dt} - \frac{1}{C}\frac{dq}{dt} - L\frac{d^2i}{dt^2} - \frac{3R^2}{2}\frac{d\Omega}{dt} = 0$$

$$-\frac{1}{C}i - L\frac{d^2i}{dt^2} - \frac{BR^2}{2}\alpha = 0 \dots\dots(1)$$



$$\tau = iRB\frac{R}{2} = \frac{iR^2B}{2}$$

Now force on hemisphere $F = \int P dA$



$$F = \int_{r=0}^{r=R} \frac{4\pi G \rho^2}{3} dr \pi r^2$$

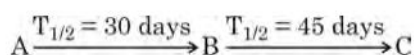
$$F = \frac{4\pi^2}{3} G \rho^2 \int_0^R r^3 dr = \frac{4\pi^2 G \rho^2}{12} R^4$$

$$F = \frac{4\pi^2 G}{12} \frac{M^2}{\left(\frac{4}{3}\pi R^3\right)^2} R^4 \Rightarrow F = \frac{4\pi^2 G}{12} \frac{M^2}{\frac{16}{9}} \frac{R^4}{\pi^2 R^2}$$

$$F = \frac{3}{16} \frac{GM^2}{R^2}$$

7. **Ans (9)**

The radioactive decay series is given



Initially $N_A(0) : N_B(0) = 3 : 1$

$$\frac{dN_A}{dt} + \lambda_A N_A = 0$$

$$\frac{dN_B}{dt} = \lambda_A N_A - \lambda_B N_B$$

$$\frac{dN_B}{dt} = \lambda_A N_A - \lambda_B N$$

$$\frac{dN_C}{dt} = \lambda_B N_B$$

$$\text{Initially } N_A = N_A(0)e^{-\lambda_A t}$$

$$N_B = c_1 e^{-\lambda_B t} + \frac{\lambda_A N_A(0) e^{-\lambda_A t}}{-\lambda_A + \lambda_B}$$

$$\text{If } N_A(0) = \frac{3}{4} N_0, N_B(0) = \frac{1}{4} N_0$$

$$\text{then we get, } c_1 = \frac{5}{2} N_0$$

$$\therefore N_A(t) = \frac{3}{4} N_0 \left(\frac{1}{2}\right)^{\frac{t}{T_{1/2}}} = \frac{3}{4} N_0 \left(\frac{1}{2}\right)^{\frac{t}{30 \text{ days}}}$$
 and,

$$N_B(t) = \left[\frac{5}{2} N_0 \left(\frac{1}{2}\right)^{\frac{t}{45 \text{ days}}} - \frac{9}{4} N_0 \left(\frac{1}{2}\right)^{\frac{t}{30 \text{ days}}} \right]$$

$$\text{Now, } \frac{\lambda_A N_A}{\lambda_B N_B} = \frac{9}{22} \text{ i.e. } \frac{N_A}{N_B} = \frac{3}{11}$$

$$\text{or, } \left(\frac{1}{2}\right)^{\frac{t}{90}} = 2$$

$$\text{or, } t = 90 \text{ days}$$

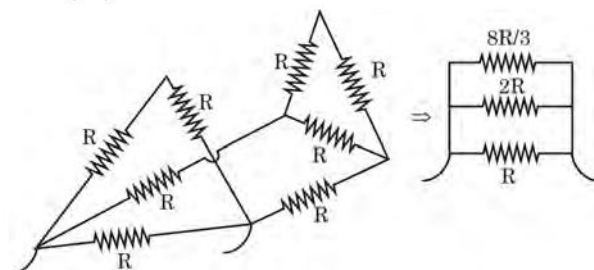
8. **Ans (3)**

$$\sigma \pi R^2 T_0^4 = \sigma 2\pi R^2 T^4 + \sigma \pi R^2 T^4$$

$$T_0^4 = 3T^4 \Rightarrow T = \left(\frac{1}{3}\right)^{1/4} T_0$$

$$T = 10(3)^{3/4}$$

9. **Ans (8)**



$$\frac{1}{R_{eq}} = \frac{1}{R} + \frac{1}{2R} + \frac{3}{8R}$$

$$\Rightarrow \frac{1}{R_{eq}} = \frac{8+4+3}{8R} = \frac{15}{8R}$$

$$R_{eq} = \frac{8R}{15}$$

10. **Ans (114)**

$$\tan i = \frac{4}{3} \therefore i = 53^\circ$$

$$\therefore r = 37$$

$$\tan i_2 = \frac{3/2}{4/3} = \frac{9}{8}$$

$$\therefore i_2 = 48.4$$

$$\alpha + 90 + r + 90 - i_2 = 180$$

$$\alpha = i_2 - 37 = (11.4)^\circ$$

PART-2 : CHEMISTRY

SECTION-I (i)

1. **Ans (B)**

$$\Delta G^\circ = \Delta H^\circ - T \Delta S^\circ$$

$$\text{Slope} = -\Delta S^\circ$$

2. **Ans (D)**

$$P.E. = \frac{kq_1 q_2}{r} = -\frac{1}{4\pi\epsilon_0} \times \frac{3e \times e}{a_0 \times \frac{(2)^2}{3}}$$

PART-2 : CHEMISTRY

SECTION-I (ii)

5. **Ans (B,C)**



$$\text{At pH} = 3, E = 1.33 - \frac{0.06}{6} \log \frac{1}{(10^{-3})^{14}} = 0.91 \text{ V}$$

$$\text{At pH} = 0, E = 1.33 - \frac{0.06}{6} \log \frac{1}{(1)^{14}} = 1.33 \text{ V.}$$

PART-2 : CHEMISTRY

SECTION-II (i)

1. **Ans (2.60)**

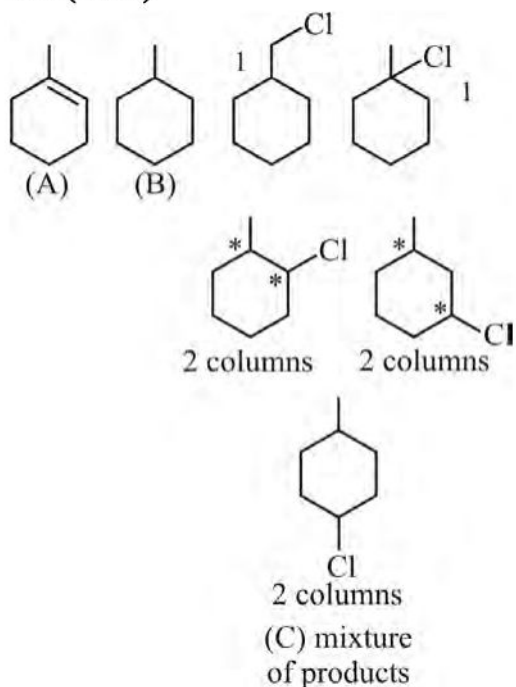
$$\text{Rate law} = k^1 [\text{CHCl}_3] [\text{Cl}_2]^{1/2}$$

$$k^1 = k_2 \left(\frac{k_1}{k_{-1}} \right)^{1/2} = 1.3 \times 10^{-2} \times \left(\frac{4.8 \times 10^3}{1.2 \times 10^3} \right)^{1/2} \\ = 2.6 \times 10^{-2} \text{ M}^{-1/2} \text{ sec}^{-1}.$$

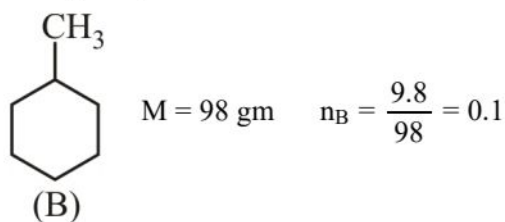
2. **Ans (42.50)**

$$E_{\text{overall}} = E_{a_2} + \frac{1}{2} (E_{a_1} - E_{a_{-1}}) \\ = 45 + 0.5 (10 - 15) = 42.5 \text{ kJ/mol.}$$

3. **Ans (8.00)**



4. **Ans (7.10)**



$$n_B = n_{\text{Cl}_2} = 0.1$$

$$W_{\text{Cl}_2} = 0.1 \times 71 = 7.10 \text{ gm}$$

PART-2 : CHEMISTRY

SECTION-II (ii)

5. **Ans (300)**

$$K_{\text{SP}} = [\text{M}^{y+}]^x [\text{SO}_4^{2-}]^y$$

$$\log K_{\text{SP}} = x \log [\text{M}^{y+}] + y \log [\text{SO}_4^{2-}]$$

From plot

$$\log [\text{M}^{y+}] = \frac{\log K_{\text{SP}}}{x} + \frac{y}{x} \log [\text{SO}_4^{2-}]$$

$$\frac{y}{x} = \frac{2}{4} = \frac{1}{2}$$

$$x = 2 \text{ and } y = 1$$

$$K_{\text{SP}} = [10^{-2}]^2 \times [10^{-4}] = 10^{-8} \text{ M}^3$$

$$z = \frac{x+y}{K_{\text{SP}} \text{ of } \text{M}_x(\text{SO}_4)_y} = \frac{2+1}{10^{-8}} = 300 \times 10^6$$

10. **Ans (2)**

v, viii are basic amino acids, which exist as cation at pH = 7

PART-3 : MATHEMATICS

SECTION-I (i)

1. **Ans (C)**

$$\text{Let } A = \lim_{n \rightarrow \infty} \frac{[(n^2 + 1^2)(n^2 + 2^2) \dots (n^2 + n^2)]^{\frac{m}{n}}}{n^{2m}}$$

$$\Rightarrow \ln A = \lim_{n \rightarrow \infty} \frac{m}{n} \sum_{r=1}^n \ln(n^2 + r^2) - 2m \ln n$$

$$= \lim_{n \rightarrow \infty} m \left[\frac{\sum_{r=1}^n \ln(n^2 + r^2) - 2n \ln n}{n} \right]$$

$$\Rightarrow \ln A = m \left[\ln 2 + \frac{\pi}{2} - 2 \right]$$

$$\Rightarrow A = e^{m \left[\ln 2 + \frac{\pi}{2} - 2 \right]}$$

$$= \left[2 \cdot e^{\frac{\pi}{2}} \cdot e^{-2} \right]^m$$

$$A = \left[\frac{2\sqrt{e^\pi}}{e^2} \right]^m \therefore a = 2$$

2. **Ans (D)**

$$\sum_{k=0}^{1012} \left[{}^{1012}C_k \sum_{r=k}^{2024} \frac{{}^r C_k \cdot {}^{2024}C_r}{{}^{2024}C_k} \right]$$

$$= \sum_{k=0}^{1012} {}^{1012}C_k \cdot 2^{2024-k}$$

$$= 2^{2024} \sum_{k=1}^{1012} {}^{1012}C_k \cdot \left(\frac{1}{2} \right)^k$$

$$= 2^{1012} \cdot 3^{1012}$$

$$= 6^{1012}$$

$$= (5 + 1)^{1012}$$

$$= 25\lambda + 5061$$

$$= 25\mu + 11$$

$$\text{Remainder} = 11$$

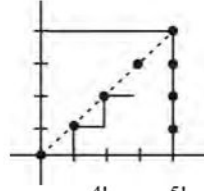
3. **Ans (A)**

(1) 3 move diagonally

(2) 2 move diagonally, 1 left, 1 down

(3) 1 move diagonally, 2 left, 2 move down

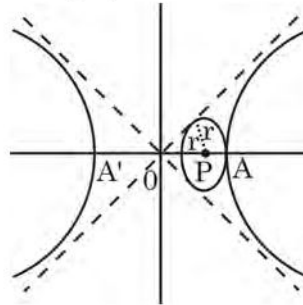
(4) 3 move left, 3 move down



$$\frac{1}{3^3} + \frac{4!}{3^4} + \frac{5!}{3^5} + \frac{6!}{3^6} = \frac{245}{3^6}$$

$$(1, 1) \text{ to } (0, 0) \rightarrow \frac{1}{3} \cdot \frac{245}{3^6} \Rightarrow \frac{245}{3^7} \Rightarrow \frac{x}{3^y}$$

4. **Ans (B)**



$$OP = r \operatorname{cosec} \theta$$

$$OA = a = r + r \operatorname{cosec} \theta$$

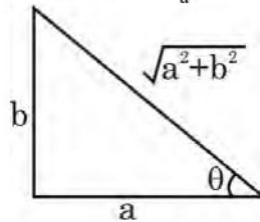
$$\Rightarrow r = \frac{a}{1 + \operatorname{cosec} \theta}$$

$$= \frac{a}{1 + \frac{\sqrt{a^2+b^2}}{b}}$$

$$= \frac{b}{\frac{b}{a} + \sqrt{\frac{a^2+b^2}{a^2}}}$$

$$= \frac{b}{\frac{b}{a} + \sqrt{1 + \frac{b^2}{a^2}}}$$

$$= \frac{b}{\frac{b}{a} + \sqrt{1 + \frac{b^2}{a^2}}}$$



$$r = \frac{b}{\sqrt{e^2-1} + e} = \frac{b(e - \sqrt{e^2-1})}{e^2 - (e^2-1)}$$

$$\Rightarrow \frac{r}{b} = e - \sqrt{e^2-1}$$

PART-3 : MATHEMATICS

SECTION-I (ii)

5. **Ans (A,B,C,D)**

$$3AABB + 3AB = 2ABAB + 2AB$$

$$\Rightarrow 3A(AB + I)B = 2A(BA + I)B$$

$$\Rightarrow 27 \|A\| \|AB + I\| \|B\| = 8 \|A\| \|BA + I\| \|B\|$$

$$\text{as } \|I + AB\| = \|I + BA\| \neq 0$$

$$\text{we get } \|B\| = 0$$

6. **Ans (B,D)**

$$f(x) = \lim_{n \rightarrow \infty} \frac{\cot x + 1}{\cot x - 1} \times \frac{(\cot x)^{2^{n+1}} - 1}{(\cot x)^{2^{n+1}} + 1}$$

$$\forall x \in \left(0, \frac{\pi}{2}\right] - \left\{\frac{\pi}{4}\right\}$$

$$\therefore f(x) = \frac{\cot x + 1}{\cot x - 1}, x \in \left(0, \frac{\pi}{4}\right)$$

$$= \frac{\cot x + 1}{1 - \cot x}, x \in \left(\frac{\pi}{4}, \frac{\pi}{2}\right)$$

$$\text{and } f\left(\frac{\pi}{4}\right) = 2, f(0) = 0$$

$$\begin{aligned} \text{Now } f'(x) &= \frac{2}{(\cot x - 1)^2} \cdot \cos \sec^2 x, x \in \left(0, \frac{\pi}{4}\right) \\ &= \frac{-1}{(\cot x - 1)^2} \cos \sec^2 x, x \in \left(\frac{\pi}{4}, \frac{\pi}{2}\right) \end{aligned}$$

$$\therefore f(x) \text{ is increasing in } \left(0, \frac{\pi}{4}\right) \text{ and } f(x) \text{ is}$$

$$\text{decreasing in } \left(\frac{\pi}{4}, \frac{\pi}{2}\right).$$

Also $f(x)$ has three local minima at $x = 0$,

$$x = \frac{\pi}{4}, x = \frac{\pi}{2} \text{ and } f(x) \text{ does not have local}$$

maxima

7. **Ans (B,D)**

$$z = e^{i\pi/13} \Rightarrow z^{13} = -1$$

$$z^{13} + 1 = 0 \Rightarrow \frac{1 + z^{13}}{1 + z} = 0$$

$$\Rightarrow 1 - z + z^2 - z^3 + \dots + z^{12} = 0 \quad \dots(1)$$

$$1 - z^{13} = 2$$

$$\Rightarrow \frac{1 - z^{13}}{1 - z} = \frac{2}{1 - z}$$

$$\Rightarrow 1 + z + z^2 + z^3 + \dots + z^{12} = \frac{2}{1 - z} \quad \dots(2)$$

from (1) & (2)

$$1 + z^2 + z^4 + z^6 + z^8 + z^{12} = \frac{1}{1 - z} \quad (\text{B option})$$

$$\& z + z^3 + z^5 + \dots + z^{11} = \frac{1}{1 - z} \quad (\text{D option})$$

PART-3 : MATHEMATICS

SECTION-II (i)

1. **Ans (2.00)**

Let the reflection of $2x + y + z - 5 = 0$ about

$$x - y + 2z + 5 = 0$$

be plane $(2x + y + z - 5) + k(x - y + 2z + 5) = 0$

$$\text{Now, } \left| \frac{(2+k) - (1-k) + 2(1+2k)}{\sqrt{(2+k)^2 + (1-k)^2 + (1+2k)^2} \cdot \sqrt{6}} \right| = \left| \frac{2-1+2}{\sqrt{6} \cdot \sqrt{6}} \right|$$

$$\Rightarrow 6k + 3 = 3\sqrt{k^2 + k + 1}$$

$$\Rightarrow k = -1 \text{ OR } k = 0 \text{ (reject)}$$

$$\therefore \text{resultant plane is } x + 2y - z - 10 = 0$$

So λ is equal to minimum distance of

$$x + 2y - z - 10 = 0 \text{ from any point on } x^2 + y^2 = 5$$

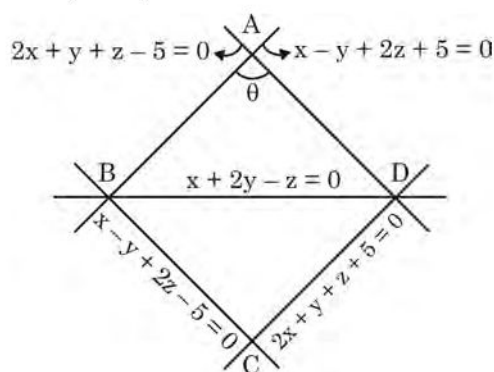
Let a point on circle be

$$(\sqrt{5} \cos \theta, \sqrt{5} \sin \theta, 0)$$

$$\text{Then } \perp \text{ distance} = \left| \frac{\sqrt{5} \cos \theta + 2\sqrt{5} \sin \theta - 10}{\sqrt{6}} \right|$$

$$\therefore \lambda = \frac{5}{\sqrt{6}} \Rightarrow [\lambda] = 2$$

2. Ans (9.00)



Perpendicular distance between opposite sides

of ABCD is $\frac{10}{\sqrt{6}}$

$$\begin{aligned}\therefore \text{Area of cross section} &= \frac{1}{2} \times \frac{10}{\sqrt{6}} \times \frac{10}{\sqrt{6}} \times \cos \theta \\ &= \frac{1}{2} \times \frac{10}{\sqrt{6}} \times \frac{10}{\sqrt{6}} \times \frac{2}{\sqrt{3}} \\ &= \frac{50}{3\sqrt{3}}\end{aligned}$$

$$\therefore [A] = 9$$

3. Ans (67.00)

$$\text{Consider } I = \int_0^1 (1-x)^m x^n dx, \text{ where } m, n \in \mathbb{N}.$$

$$\text{Put } x = \sin^2 \theta \Rightarrow dx = 2 \sin \theta \cos \theta d\theta$$

$$\Rightarrow I = 2 \int_0^{\frac{\pi}{2}} (\cos \theta)^{2m+1} (\sin \theta)^{2n+1} d\theta$$

$$\Rightarrow I = \frac{2 \times (2m)(2m-2) \times \dots \times 2 \times (2n)(2n-2) \times \dots \times 2}{(2m+2n+2)(2m+2n) \times \dots \times 2}$$

$$\Rightarrow I = \frac{m!n!}{(m+n+1)!}$$

$$\therefore I_1 = \frac{6!4!}{11!}, I_2 = \frac{5!3!}{9!}$$

$$\Rightarrow \frac{I_1}{I_2} = \frac{6 \times 4}{11 \times 10} = \frac{12}{55}$$

4. Ans (22.00)

$$I = \frac{1}{2^5} \int_0^{\frac{\pi}{4}} \frac{(\cos x - \sin x)^{10}}{(\cos x)^{10}} \times \frac{(\sin x)^7}{(\cos x)^7} \cdot \sec^2 x dx$$

$$\tan x = t$$

$$I = \frac{1}{2^5} \int_0^1 (1-t)^{10} \cdot t^7 dt$$

$$I = \frac{10!7!}{2^5 \times 18!}$$

PART-3 : MATHEMATICS

SECTION-II (ii)

5. Ans (3)

$$(2xy - y)(x^4 + x^2 + 2) = 0$$

$$\Rightarrow y = 0; x = \frac{1}{2}$$

$$\text{Also, } x^3 + y^4 = x$$

$$y = 0; x = -1, 0, 1$$

$$x = \frac{1}{2}; y = \pm \left(\frac{3}{8}\right)^{1/4}$$

6. Ans (0)

$$(2x+1)^2 + (2y-1)^2 = 8086$$

$$x, y \in \mathbb{I} \quad 8k_1 + 1 + 8k_2 + 1 = 8086$$

$$8(k_1 + k_2) = 8084$$

not possible

7. Ans (80)

$$\text{Let } x_1 = 3.$$

$$3(x_2x_3 + x_4x_5 + x_2x_5) + x_3x_4(x_2 + x_5)$$

$$x_2 + x_5 \text{ must be } 3\lambda$$

$$\text{i.e., } (1, 2), (1, 5), (2, 4), (4, 5) \rightarrow 8 \text{ ways}$$

$$\text{Req. ways} = 5 \times 8 \times 2 = 80$$

8. Ans (4)

$$\text{At } x = 0 \quad \frac{-k}{(-2)^4} + \frac{6}{(-1)^4} = 0 \Rightarrow k = 96$$

$$(\sin \theta + 2 \cos \theta)(\sin \theta + \cos \theta) = 0$$

$$\tan \theta = -1, -2$$

9. Ans (4)

$$A^2 = 7A$$

$$\Rightarrow \text{tr}(A^n) = 0^n + 7^n = 7^n$$

$$\text{i.e., } n = 4$$

10. Ans (5)

$$\frac{x^2}{100} + \frac{y^2}{225} = 1; (x-2)^2 + (y-3)^2 = 25$$

$$(y-3)^2 = 6(x-7)$$

